

G E O G R A P H Y

INSTANT LESSONS POWERPOINT

OUR PHYSICAL WORLD



Your PowerPoint CD

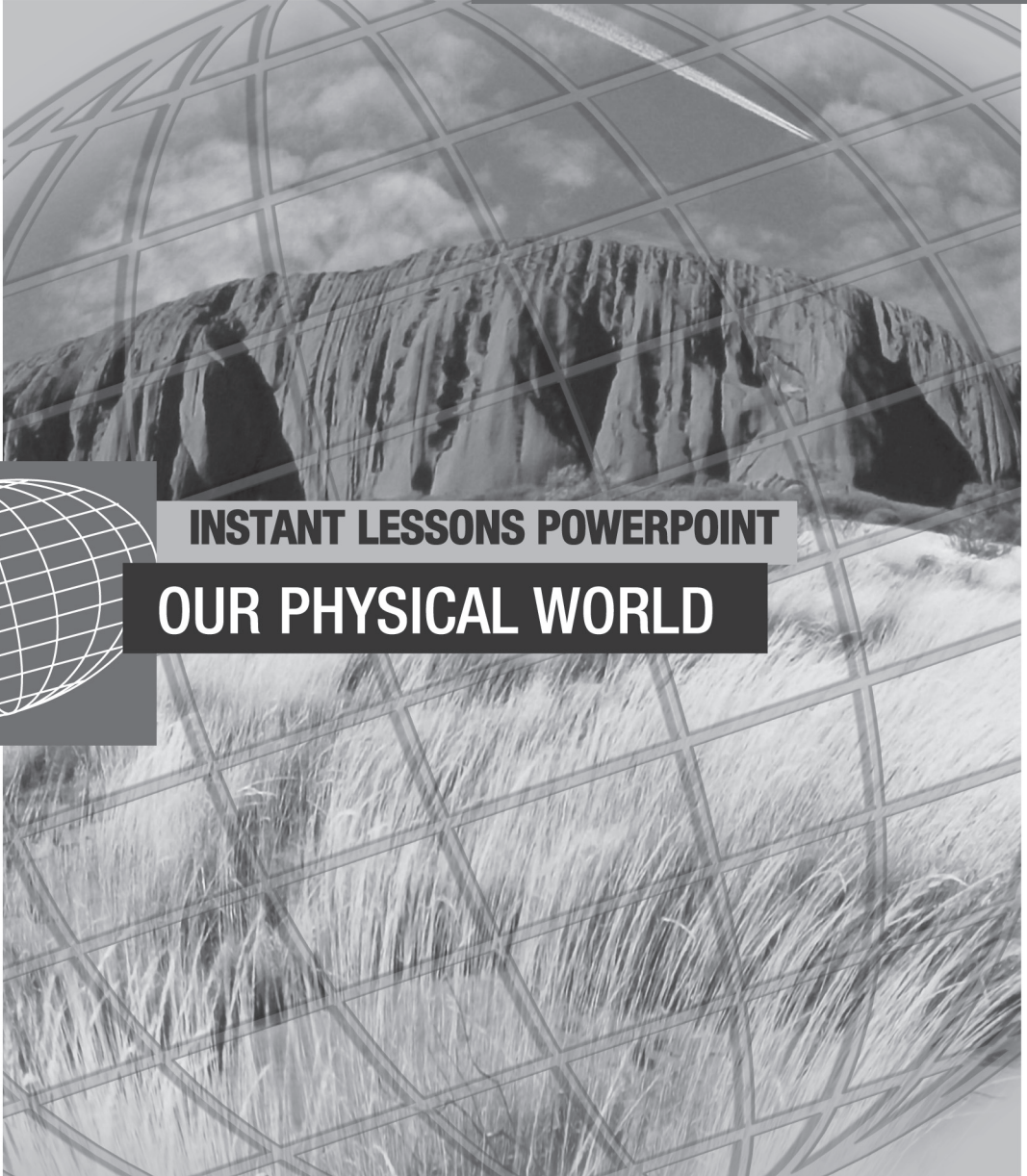
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GEOGRAPHY



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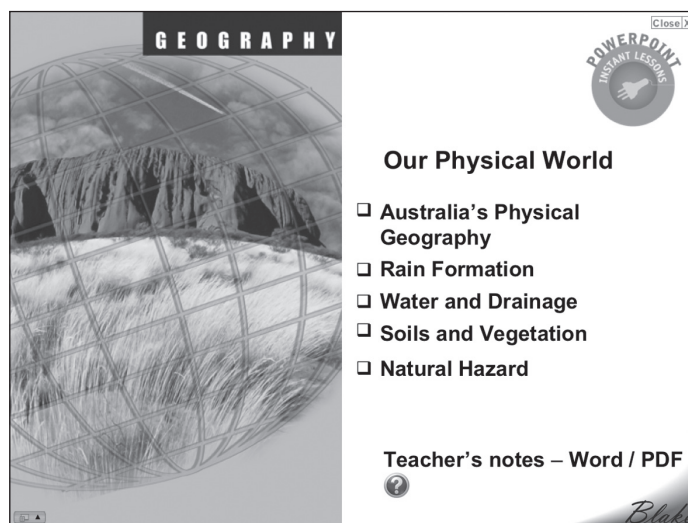
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INTRODUCTION

Instant Lessons PowerPoint slide presentations have been designed to assist teachers in presenting essential geography curriculum to junior and middle secondary school students. Physical, environmental, social and political geography themes are included throughout the series.

The enclosed CD-ROM contains:

- 5 PowerPoint slide presentations
- this book, in PDF format
- the activity sheets in this book, as editable word documents.



Key features and uses:

- Each topic is explored through a core set of slides.
- Teachers can tailor-make their own slide presentations by adding, deleting or altering slides.
- Carefully chosen diagrams, maps, graphs, illustrations and photographs illustrate the concepts being taught.
- On-screen activities test and extend student comprehension.
- Slide notes are provided as a 'map' for the content of each slide.
- Teacher's notes are provided in this book for each topic.
- Student worksheets have been designed so that they can be used during the presentation or later in the classroom, or as homework, extension, assessment or revision. They can be accessed on the CD-ROM in PDF and Word formats, or photocopied from this book.

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It is recommended that teachers preview the slide sets before showing them to their class, to become familiar with the format and features.

Read the Teacher's notes for an overview of each topic, including key terms and concepts.

The PowerPoint presentations

Starting a slideshow presentation

The Slide Show is used to display the presentation on the screen. It is recommended that you use an external projector for whole-class use.

There are a number of ways to begin a Slide Show presentation. In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click on the presentation you wish to view.

In the Macintosh environment, double-click the file index.pps. From the Main menu, click on the presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

Navigating

Main menu

- ◆ From the Main menu, click Close to quit.
- ◆ During the presentations, click Close in the top right corner at any time to return to the Main menu.

Navigating use the mouse

- ◆ Click – Moves presentation to the next slide.

Moving forward

- ◆ Press the N key.
- ◆ Press the Spacebar or Enter key.
- ◆ Press the Page Down key.
- ◆ Press the Right Arrow key.

Moving backward

- ◆ Press the P key.
- ◆ Press the Backspace key.
- ◆ Press the Page Up key.
- ◆ Press the Left Arrow key.

Navigating using the slide show toolbar

At the bottom left corner of the slide show, a toolbar is displayed. There are three buttons that deal with navigation.

- ◆ Left Arrow button – Moves presentation back to the previous slide.
- ◆ Right Arrow button – Moves the presentation to the next slide
- ◆ Slide Icon button – Provides a pop-up menu with Next and Previous menu items, as well as a Go to Slide item that allows for navigation to a specific slide.
- ◆ Right-Click – Provides a pop-up menu with Next and Previous menu items, as well as a Go to Slide item that allows for navigation to a specific slide.

Navigating to a specific slide

There are a number of different ways to display a specific slide.

- ◆ Slide Number + Enter – While in the Slide Show mode, type in the slide number to be displayed and press Enter.
- ◆ Control-S – To display a menu of all slides in the presentation, press Control and S keys. A menu will appear. Select the desired slide from the menu and press Enter.
- ◆ Slide Icon Button – Choose the Slide Icon button from the Slide Show Toolbar. Choose Go to Slide, then choose the desired slide.
- ◆ Right-Click – Choose Go to Slide, then choose the desired slide.

Blanking out the screen

To present a blank screen while in the Slide Show mode:

- ◆ Slide Icon Button – Choose the Slide Icon button from the Slide Show Toolbar. Choose Screen menu item. Select Black Screen or White Screen.
- ◆ Right-Click – Choose Screen menu item. Select Black Screen or White Screen.

Speaker notes

Speaker notes are included for each slide to provide a summary of the content of that slide. In Normal view, these notes appear at the bottom of each slide.

Speaker notes can be displayed during a slide show presentation. However, when displayed they will be visible to the audience. If you do not want students to see the speaker notes, but need to access them, print the notes before class from the Word document on the CD-ROM.

To view speaker notes during a slide show presentation:

- ◆ Right-click, choose Screen | Speaker Notes to see the notes for this slide.

Pointer / pen

The Pointer and Pen tools can be used in the Slide Show view to highlight or draw on the screen, much as would be done on a whiteboard.

Drawings can be saved with the presentation or discarded.

Note that while using Pointer tools during a Slide Show, you will not be able to click the mouse to move to the next slide. Instead, use one of the alternative navigations provided on the previous page, such as the Right Arrow key.

To activate the Pointer tools:

1 In Slide Show view, right-click, choose Screen | Pointer Options (or click on the Pen icon in the Slide Show Toolbar) and choose one of the following.

- ◆ Arrow – provides an arrow on the screen.
- ◆ Ballpoint Pen – used to write notes on the screen.
- ◆ Felt Tip Pen – a wider pointed pen, useful for drawing. (Windows Office 2003 or higher only)
- ◆ Highlighter – used to apply a highlight of color that is transparent. (Windows Office 2003 or higher only)
- ◆ Ink Color – Choose the color to be used with the tool.
- ◆ Eraser – used to erase entries. (Windows Office 2003 or higher only)
- ◆ Erase All Ink on Slide – Removes all entries from the screen. (Windows Office 2003 or higher only)

2 Use the mouse to draw or write on the screen.

Ending a presentation

Each of these actions will end the presentation:

- ◆ Click Close in the top right corner.
- ◆ Press the Esc key.
- ◆ Right-click and choose End Show.
- ◆ Choose the Slide Icon button in the Slide Show Toolbar & select End Show.

Technical specifications Windows

The CD-ROM will autorun in the Windows environment.

To view the presentations – Windows 97 or later operating system

To edit a presentation, open it from within the PowerPoint application.

Note that if editing these PowerPoint presentations in a version prior to Office 2000, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

Double-click the file index.pps on the CD-ROM. From the Main menu, click on the presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed. (i.e. Office 2004 for Macintosh or later)

To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Teacher's notes on the CD-ROM

To open these files you need:

- ◆ **PDF file** – Adobe Acrobat Reader version 6, or higher.
- ◆ **Word document** – Microsoft Office 2000, or higher.

TOPIC 1 – Australia’s Physical Geography

These slides look at how the Australian land was formed and the particular forces that shaped it. Students will gain an understanding of the range of landforms and land types in Australia, as well as learn about gradational forces. Australia’s climatic regions are also examined, including the air masses that affect the continent and common patterns of weather in summer and winter. This title also includes extension case study slides about Uluru and Kata Tjuta and the natural forces that helped create these well-known landforms.

- Key terms:** Gradational and tectonic forces, erosion, weathering, folding, faulting, vulcanism, deposition, climatic regions, air masses
- Concepts explored:** Australia – a stable continent; types of landforms and how they are created – residual, erosion, deposition; landform features and how they were formed; Australia’s landform regions; facts and features about Australia quiz; Australia’s climate; air masses that affect Australia; climatic regions in Australia; weather patterns; typical winter and summer conditions
- Examples used:** Australian and regional examples such as the Flinders Ranges, the Kimberley region, Mount Gambier, Ningaloo Reef, Kangaroo Island, Eastern Highlands, Melville Island, Mt Kosciuszko, Wave Rock
- Extension slide sets:** Case study – Uluru and Kata Tjuta. These slides introduce the landforms Uluru and Kata Tjuta and describe their age, location and how they are part of an alluvial fan and were formed by erosion.

TOPIC 2 – Rain Formation

This set of slides introduces students to the three main ways in which rain is formed – convection, orographic and frontal. The emphasis is on understanding the differences between each kind, where they commonly occur and how this impacts on land use and water collection.

- Key terms:** Convection, orographic, frontal
- Concepts explored:** How water is used; types of precipitation; temperature; local conditions; windward and leeward; types of landforms; vapour; evaporation; rainshadow; land use and water use; average and annual rainfall; urban and non-urban land use; density, mass and volume
- Examples used:** The Macedon Ranges, Victoria; The Daintree Forest, Queensland; The Blue Mountains, New South Wales
- Extension slide sets:**

Set 1 – Land use and available water. Students are required to use what they know about how rain is formed to locate a dam to collect water for urban use.

Set 2 – Measuring the density of water and air, and showing how this relates to air movement and rain formation.

TOPIC 3 – Water and Drainage

- Key terms:** The water (hydrologic) cycle; groundwater and runoff; river systems; global warming
- Concepts explored:** Sources of water; key parts of the water cycle; collecting water from groundwater runoff; the river system and water collection; state variations in water use and resources; water use and population; climatic and human pressures on water supplies; household water use – a global perspective
- Examples used:** State and international figures on water use – United States of America, Canada, United Kingdom, France, Israel, Germany, Bangladesh, Sweden and South Africa

Extension slide sets: Water use in an Australian city. In this activity, students look at how water is used in households and at the measures that are often suggested to help solve water supply problems. They are asked to examine the advantages and disadvantages of each measure and to reflect on how these measures would impact on their own lifestyles.

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TOPIC 4 – Soils and Vegetation

This set of slides highlights the ancient nature of Australia’s soil and vegetation and how easily it has been degraded since white settlement through agriculture and other uses. Soil profiles are used to explore the characteristics of soil and this information will help students to understand more clearly the extension slides about salinity.

Key terms: Profile diagrams, soil, biomes, the notion of value, soil degradation, salinity

Concepts explored: Australia – an ancient continent; parts of vegetation and soil profiles, what soil is; soil characteristics; typical rainforest soils; patterns of vegetation; land clearing; measures to protect forests; events impacting on Australia’s vegetation – Rio Earth Summit, continental drift, European settlement, fire-stick farming, Australia separates from Gondwana; vegetation types in Australia; the value of forests; native forest clearing; salinity – cost to agriculture, government initiatives; agricultural regions; proportion of farms affected by salinity

Examples used: Australian statistics

Extension slide sets: Soils and salinity. These slides examine why Australian soils are so poor and how easily they can be degraded. The causes of salinity are discussed as well as the extent of the problem in Australia. Key definitions explored include salinity, leaching and fertility. Students are quizzed about their understanding of the topic and it is likely that they will need to see the whole slide set to answer the questions fully.

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TOPIC 5 – Natural Hazards

This set of slides is about different types of natural hazards and how they occur. In particular, earthquakes, volcanoes and tsunamis are described. Students will study a map of hazards in Australia and look at specific examples that have occurred in this country.

Concepts explored: Natural hazards; Earth’s crust – mantle, outer and inner core; tectonic plates and movement; parts of a volcano; tsunamis – generation, propagation and inundation; Richter Scale; community aid

Examples used: 2004 Indian Ocean tsunami, Thredbo landslide; Ash Wednesday fires; Hobart fires; Meckering earthquake; Sydney hailstorm; Cyclone Tracy; Great Influenza Epidemic; Newcastle earthquake; Southern Australian heatwave

Extension slides: Hazard risk mapping. Students will examine the places where hazards are more likely to occur in Australia and develop an understanding of why.

AUSTRALIA'S PHYSICAL GEOGRAPHY 1

Australian landforms

1 Identify these seven processes as either tectonic or gradational: *weathering, folding, faulting, erosion, vulcanism, landslide, deposition*.

Gradation	
Tectonic	

2 Write each landform beside the correct description. Then identify it as residual, erosional or depositional. Landforms: *gibber plain, beach, valley, dune, mountain, plain, canyon, delta*.

Description	Landform	Type
An area of lower land, usually between hills or mountains		
An accumulation of material at a river-mouth		
A deep, steep-sided valley		
A stony surface where wind has removed finer materials		
A large, flat, low-lying area		
Material that accumulates along a coastline		
A very high, steeply sloped landform feature		
An undulating pattern of windblown material		

3 In a journey across Australia from west to east, following the Tropic of Capricorn, the main types of landform you would cross (in order) are:

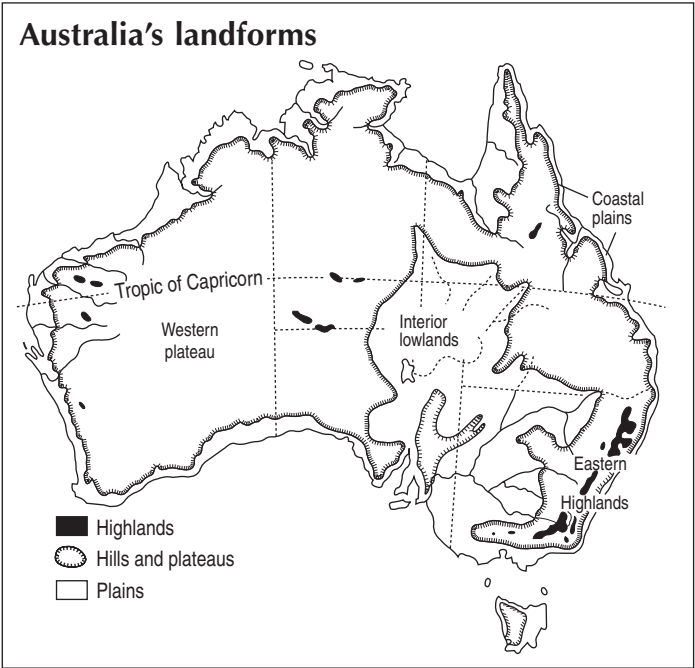
- a coastal plains, highlands, interior lowlands, hills and plateaus, coastal plains.
- b coastal plains, interior lowlands, hills and plateaus, highlands.
- c coastal plains, hills and plateaus, interior lowlands, hills and plateaus, coastal plains.
- d coastal plains, interior lowlands, coastal plains.

4 Desert landforms, such as sand dunes and canyons, would most likely be located in the:

- a highlands and interior lowlands.
- b interior lowlands and coastal plains.
- c interior lowlands and plateau.
- d western plateau and highlands.

5 Australia has ancient, extinct volcanoes in the eastern states. The volcanoes are oldest in Queensland, of intermediate age in NSW and youngest in Victoria and South Australia. This most likely provides evidence that Australia:

- a is drifting closer to New Zealand, which has active volcanoes.
- b has been drifting northward over a hotspot in the Earth's crust.
- c has been drifting southward over a hotspot in the Earth's crust.
- d is about to experience renewed volcanic eruptions.



AUSTRALIA'S PHYSICAL GEOGRAPHY 2

6 Write these landform features below the correct example: *fold mountains, volcano, block mountains, dissected plateau.*

Flinders Ranges	Kimberley	Mount Gambier	Eastern Highlands

7 Complete the sentences, using these words: *flattest, largest, driest.*

- a Australia is the world's _____ island, with a land area of about 7 692 000 km².
- b Australia is also the world's _____ continent, with its highest point measuring 2228 m.
- c Australia is the second _____ continent in the world, after Antarctica.

True or false?

8 Mark the following statements as true or false, then correct the false ones so that they are true.

- a Cape York Peninsula is north-west of Alice Springs. _____
- b Melville Island is Australia's second largest island after Tasmania. _____
- c The Flinders Ranges are located in Western Australia. _____
- d Mt Kosciuszko is Australia's highest point. _____
- e Perth's climate can be described as tropical and wet. _____

Which place am I?

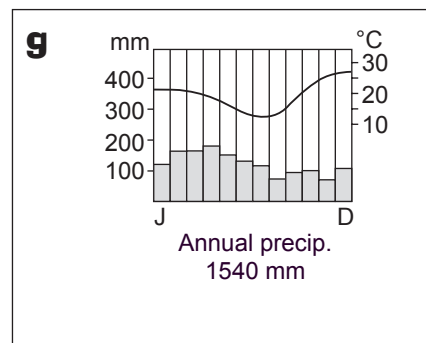
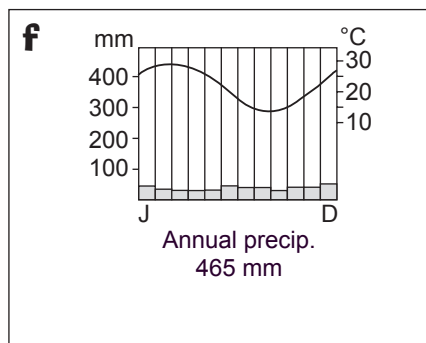
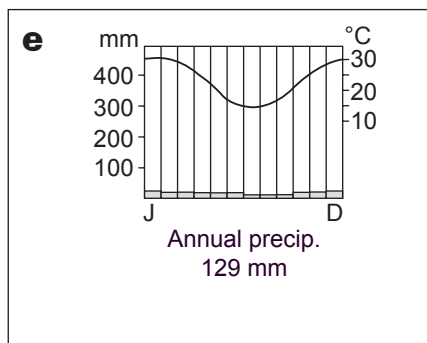
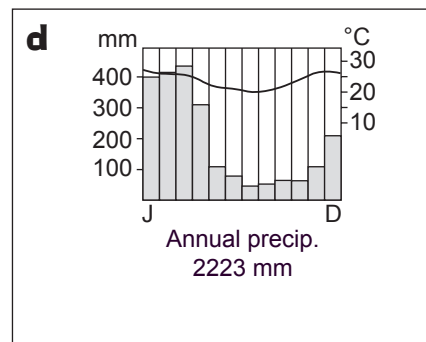
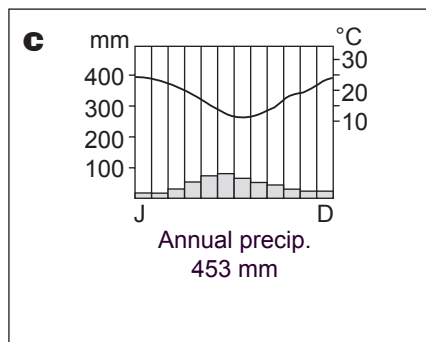
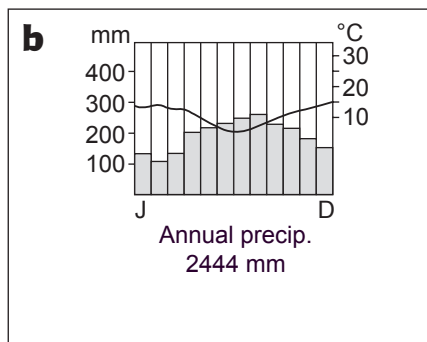
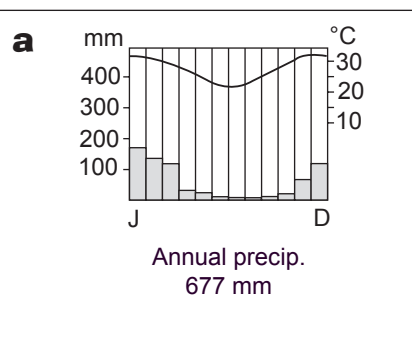
9 Identify these Australian places.

a	I am a 260-metre long reef. I am the largest fringing coral reef in Australia. I contain over 500 species of fish and 200 species of coral.	
b	I am located 110 km south-west of Adelaide. I am Australia's third largest island. My natural attractions make me very popular with tourists.	
c	I am the world's largest sand island. I am 125 km long and over 160 000 hectares in area. I was formed during the Ice Age, when wind transported vast quantities of sand from NSW and deposited it along the Queensland coast.	
d	I run 300 km east and west of Alice Springs. I am the second longest and second highest mountain range in Australia. I have red sandstone gorges and waterholes lined with red river gums and ghost gums.	
e	Scientists believe I formed 2700 million years ago. I am 14 metres high and 110 metres long. I look like I am just about to break.	
f	I am located 987 km west of Sydney. I am one of 17 dry lakes which constitute the Willandra Lakes World Heritage Area. I dried up 14 000 years ago and am now a rich source of fossils.	
g	I am a rainforest located in tropical Far North Queensland. I am 135 million years old. I was placed on the World Heritage List in 1988.	
h	I am 1545 metres tall. I am one of the most popular tourist destinations in Tasmania.	

AUSTRALIA'S PHYSICAL GEOGRAPHY 3

10 Write each climatic region below the correct climate graph.

- Tropical East Coast
- Tropical Wet-Dry Monsoon
- Tropical Desert
- Warm Temperate East Coast
- Warm Temperate West Coast (Mediterranean)
- Mid Latitude Continental
- Cool Temperate West Coast



Source: Commonwealth Bureau of Meteorology

11 Use these words to complete the sentences: *driest, deposition, sand, rock, water, pressure, exposed, alluvial, tectonic*.

Today, Uluru is located in one of the _____ parts of Australia. But it was formed 550 million years ago by the action of _____.

Erosion of sand and rock by rain from nearby mountains resulted in _____ of the material into large _____ fans. _____ from the sea, mud and sand that settled on these fans caused them to become _____. After upheaval caused by _____ forces, Uluru shifted about 90° degrees. Erosion of the softer rock has left the hard rocky surface of Uluru _____ and the surrounding area is covered in _____.

12 Research task: Uluru – The above explanation of Uluru's formation is based on modern geological knowledge. Indigenous people explain it very differently. Find out about their story of how Uluru came to be.

RAIN FORMATION 1

- 1

Life on Earth depends on water for survival. What are some ways that humans use and depend on water?
- 2

Water reaches Earth as precipitation. Define ‘precipitation’ and describe the main types.
- 3

Describe where on Earth precipitation mostly falls in solid form.
- 4

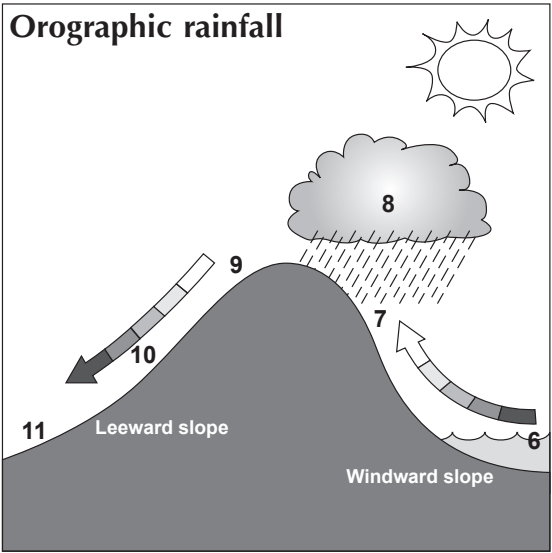
What are ‘local conditions’? List some examples.

How rain is formed

- 5

Type one: Orographic rainfall – Orographic means ‘being formed by rising over a high landform’. Write each number in the diagram beside the correct description in the table.

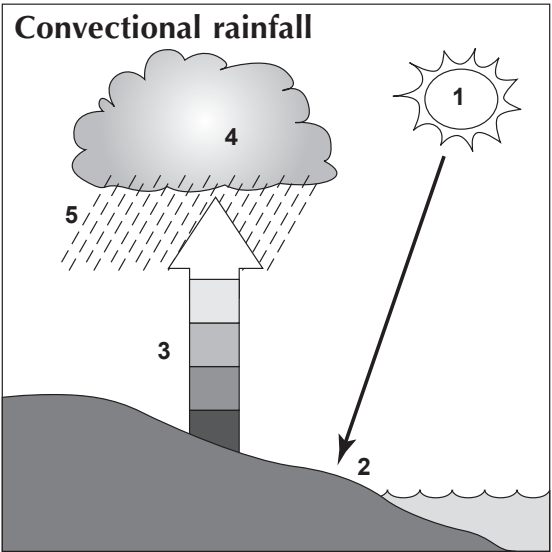
	Dry air slides down the slope and starts warming up again
	Evaporation occurs over the sea
	Water vapour in the air condenses and cools, forming rain clouds
	Cold air becomes dry
	Moist air is forced to rise over a hill or mountain, cooling at a rate of 0.6 degrees Celsius per 100 metres
	Clouds disappear and the air is dry with warm winds, creating a rainshadow.



- 6

Type two: Convectional rainfall – Convectional rain occurs when the Sun heats the ground causing air to rise rapidly. Write each number in the diagram beside the correct description in the table.

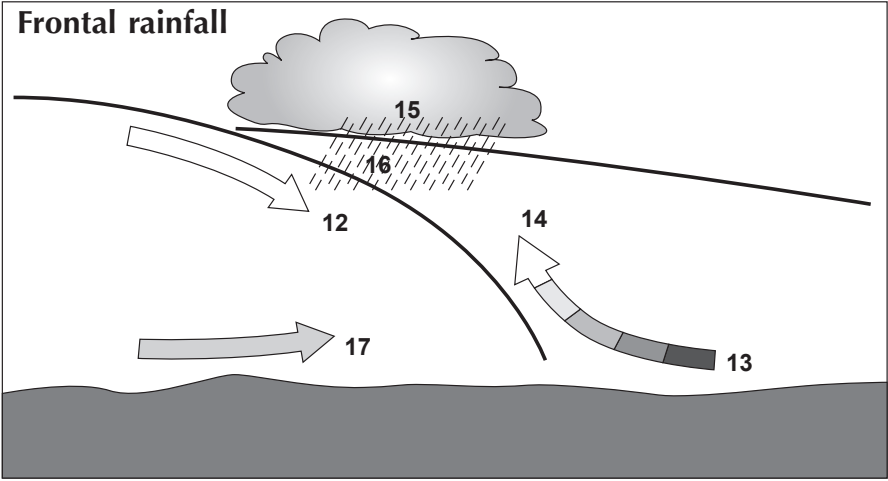
	Large clouds develop directly above the rising air
	The Sun heats up the ground rapidly
	Heavy rain and thunderstorms result, due to rapid cooling where water droplets collide with each other and fall
	The ground heats up through convection and the air rises rapidly, taking with it evaporated water
	The quickly rising air cools rapidly



RAIN FORMATION 2

7 Type three: Frontal rainfall – When large bodies of air of different temperatures meet, they form a front. They do not merge because one is warm and the other is much colder. The warmer air rises above the colder air, cools and condenses. Write each number in the diagram beside the correct description in the table.

	The air is forced to rise and it cools
	The cold air is heavy and pushes forward along the ground
	Further upward movement in clouds causes rain
	Frontal movement
	Water vapour condenses to form clouds
	The warmer air is lighter, so is forced to rise above the cold air mass.



So what do you know about air, water and rain formation?

- 8** Pick the right word in *italics* and write it into the sentence.
- a** Warm air _____. (*rises / falls*)
 - b** Warm air picks up _____. (*grass / water*)
 - c** _____ air can hold water. (*Warm / Cold*)
 - d** As air rises, it _____. (*cools / lightens*)
 - e** Air that is getting cooler _____ water. (*heats / releases*)
 - f** Water falls from _____ air. (*windy / cooling*)
 - g** Heat is released when it _____. (*rains / snows*)

Test your understanding

- 9** Read each statement and mark it as true or false. Rewrite the false statements so that they are true. Add another short extension sentence to the true statements.
- a** Rain is one type of precipitation.

 - b** Rain is not formed when air rises quickly.

 - c** A rainshadow forms on the windward slope of a mountain.

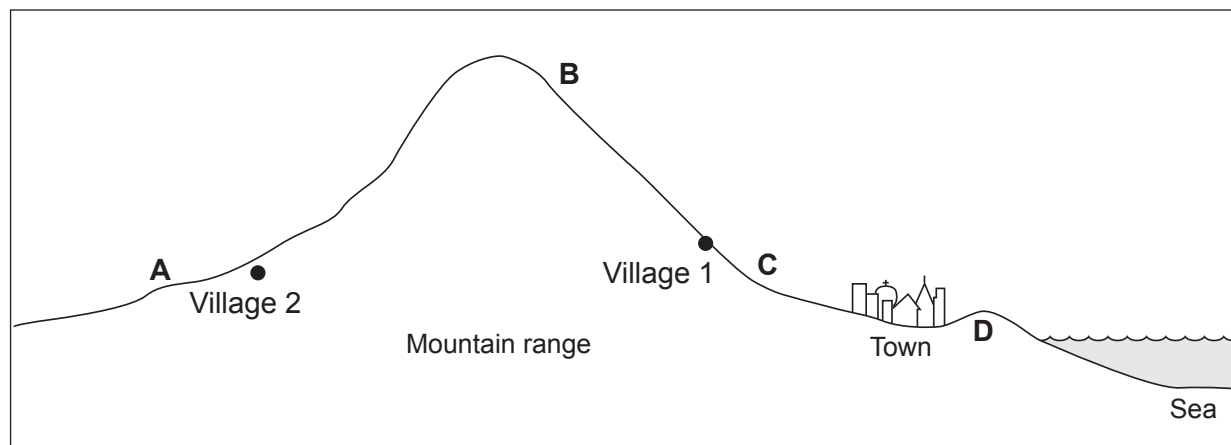
 - d** Convectional rain occurs more often in local conditions that are cool and dry.

 - e** Thunderstorms and heavy rainfall are often the result of convectional rain.

RAIN FORMATION 3

Orographic rainfall, settlement and land use

- 1** This town needs a new supply of fresh water. Which is the best location for the dam – A, B, C or D? Why?



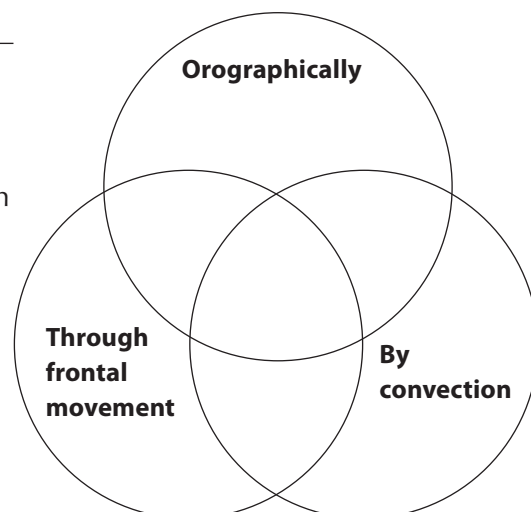
- 2** Which village gets the highest rainfall? ____
- 3** These types of agricultural produce are grown in the area: vegetables, dairy produce, olive, wheat, timber. Which type do you think is grown around which village? Give reasons, related to rainfall.

Case study: Your local area

- 4** What is the average rainfall? _____
- 5** Do environmental conditions (e.g. latitude, landform, proximity to the sea) mean that most rainfall is orographic, convectional or frontal?
- _____
- 6** What are the main non-urban land uses surrounding the city or region where you live?
- _____
- 7** Identify how rainfall and other climatic characteristics impact on those land uses.
- _____

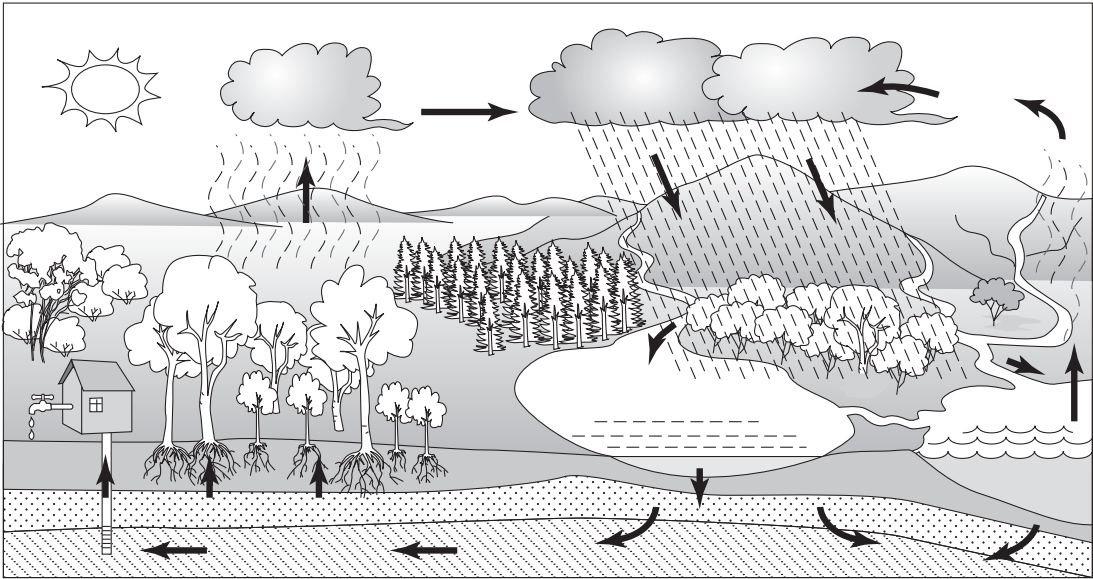
- 8 a** In your workbook, draw up a Venn diagram like the one shown here. Use it to compare and contrast the three main ways rain is formed: orographically, by convection and through frontal movement.

- b** What is the key feature that all three forms share?
- _____

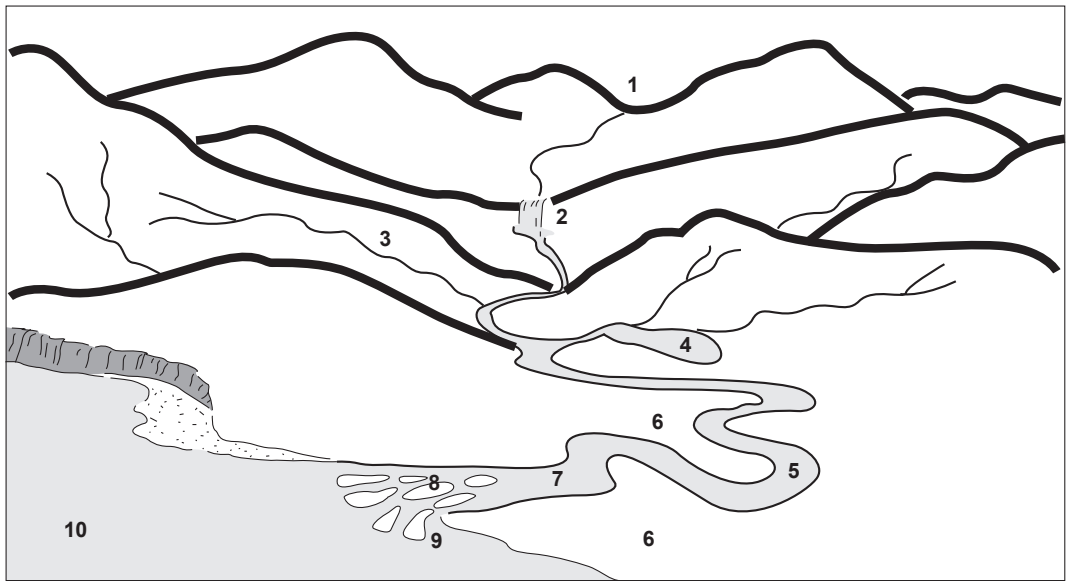


WATER AND DRAINAGE 1

- 1** Water exists on Earth's surface and in the atmosphere. It can be in a liquid, solid or gaseous state. List an example of water in each state.
-
- 2** Using the listed terms, label the water diagram. Then annotate each stage to describe what is happening in the cycle.
- Terms: *groundwater storage, precipitation, evaporation, groundwater runoff, condensation, transpiration, infiltration, surface runoff, collection.*



A river system



- 3 a** Use these terms to label the diagram above: *mouth, waterfall, estuary, tributary, sea, lake, source, flood plain, delta, meander.*
- b** Add colour and arrows to the diagram to show the flow of water in this system.
- c** Seventy-three percent of water used in Australia comes from rivers. Describe where the water you use comes from. If river systems supply your water, name the rivers.

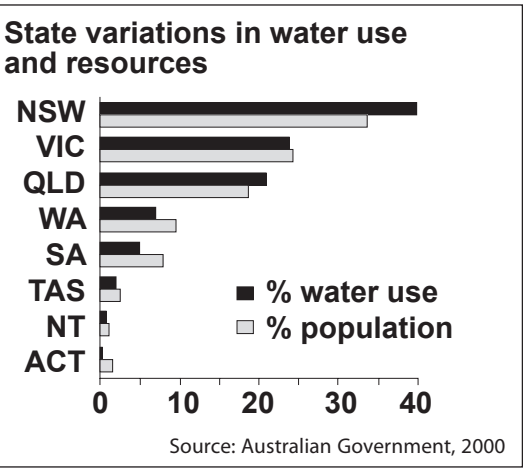
WATER AND DRAINAGE 2

4 a What does this graph show you?

b Describe the general relationship between water use and population in Australia.

c Identify the states/territories that have a higher share of water use than they do of population.

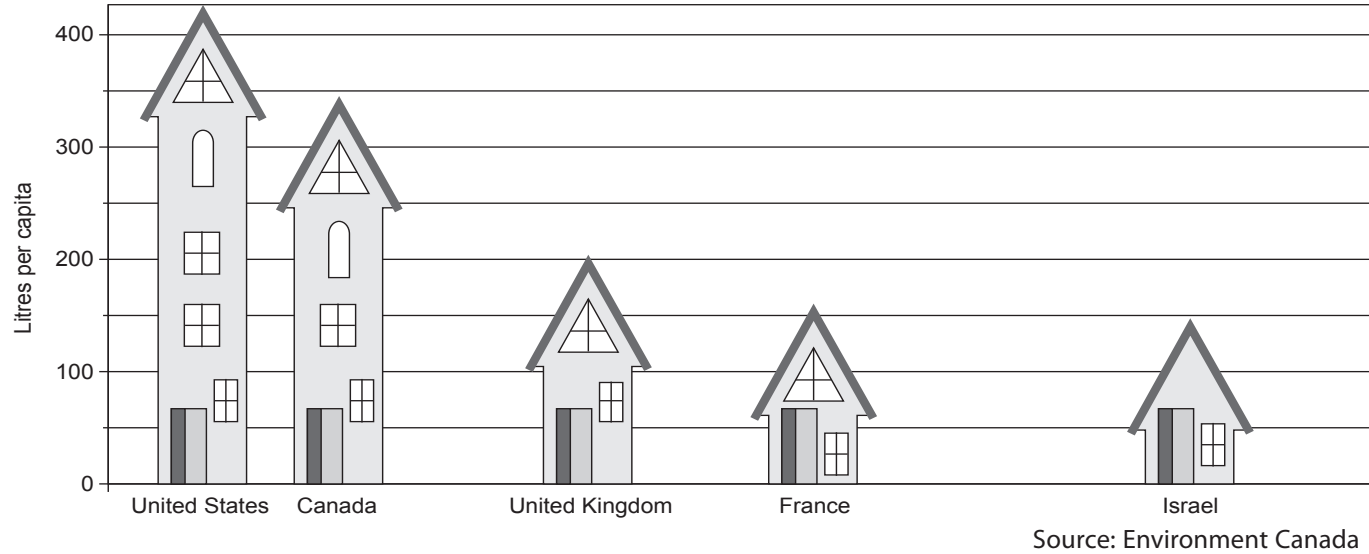
d Suggest reasons why this is the case.



e Which state/territory appears to be the most efficient user of water, per person?_____

5

Average daily household water use



a What does this graph show?_____

b What does 'litres per capita' mean?_____

c On the graph, draw the use rates for Australia (340 L), Germany (150 L), Bangladesh (40 L), Sweden (200 L) and South Africa (150 L).

d List the countries in order from largest user per capita to smallest user per capita. Include the number of litres for each country.

e What can you conclude about water usage and socio-economic levels from this data?

g Write and answer your own question about the graph and water use rates.

WATER AND DRAINAGE 3

The issue of water supply

- 1** Use research and your own knowledge to:
- a** list five measures to help solve water supply problems. (The first one has been done for you.)
 - b** list the advantages and disadvantages of each measure.
 - c** comment on how each measure could impact on you and your family. If the impact is negative, consider how you may need to change your lifestyle or attitudes in response to the realities of water supply.

Measures to solve water supply problems	Advantage	Disadvantage	Impact
Building more dams			
Banning ...			

SOILS AND VEGETATION 1

Tools for looking at vegetation and soil

1 In your own words, describe what soil or vegetation profiles look like and how geographers may use them in their work.

2 Is a profile more or less useful than a photograph as a representation of the rainforest? Why?

3 Use the rainforest profile below to help you answer these questions.

a What is a profile?

b What the profile showing?

c Why are the measurements there?

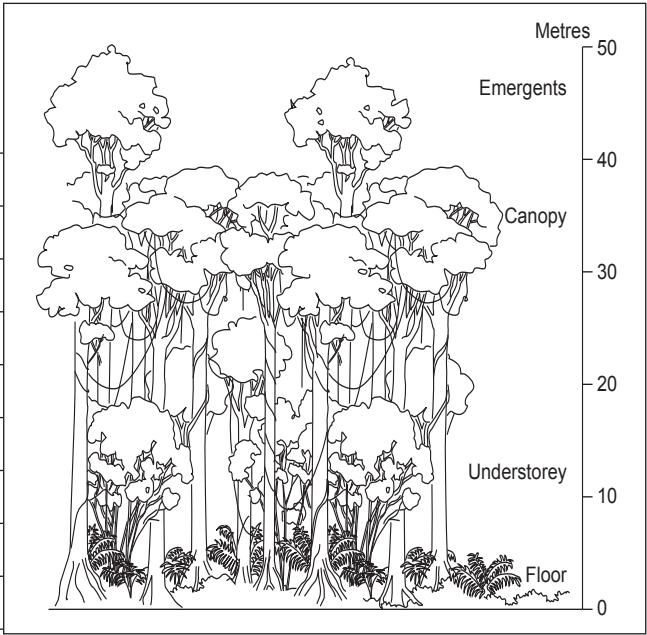
4 Look at the profile diagram below and write a definition for each rainforest layer shown. Use the scale to help you.

a Floor

b Understorey

c Canopy

d Emergents



5 Soil can be defined as:

6 Complete the table by filling in further examples of the main ingredients of soil.

Soil consists of:	Main ingredients		
Weathered rock pieces	limestone		
Organic matter	garden scraps		
Minerals		iron	
Gases		carbon monoxide	
Water (What is the source?)			domestic garden watering
Living organisms		beetles	

SOILS AND VEGETATION 2

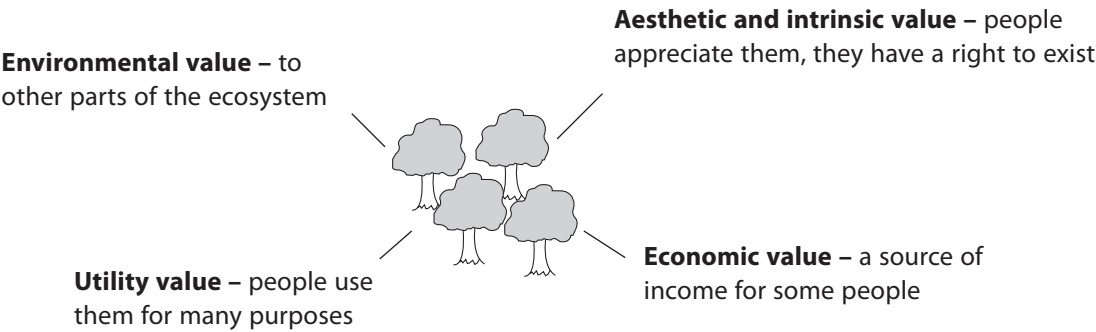
Australia’s vegetation and forest resources

7 Australia’s pattern of vegetation has developed as a result of several factors. For each factor listed below, provide an example or explanation for how it has affected vegetation patterns. The first one has been completed for you.

Geologic history	The continent of Australia was once connected to other land masses, but later became a separate continent. Over time, this isolation helped to create enormous biodiversity in plants and animals.
Past and present climate	
Location	
Size	
The impacts of its inhabitants	
Forest protection programs	

The value of forests

8 List a specific example for each value shown below. Then add at least two other ways of valuing trees and forests.



Native forests now cover just over 5% of the Australian land mass. Since European settlement, this has been reduced from about 69 million hectares to about 43 million hectares. The area of woodland has been reduced from about 157 ha to 112 million ha.

- 9 a** Calculate the % of original native forests that are still alive. _____
- b** What % has been cleared since European settlement? _____
- 10 a** Calculate the % of original woodland that is still alive. _____
- b** What % has been cleared since European settlement? _____

SOILS AND VEGETATION 3

Soils, salinity and agriculture

- 1 Use these words to complete the following sentences: *salinity, farming, minerals, topsoil, poor, irrigation, evaporation, underground*.
- Salinity is one reason why some Australian agricultural soils are so _____.
_____ can be caused by a number of factors, including:
- the _____ of water
 - water being added to soil in large volumes, as occurs with _____
 - the _____ water table rising, bringing dissolved salts and _____ upwards into the _____.
- Salinity is a problem in most of Australia's main _____ regions, and is particularly serious in the southern half of the continent.

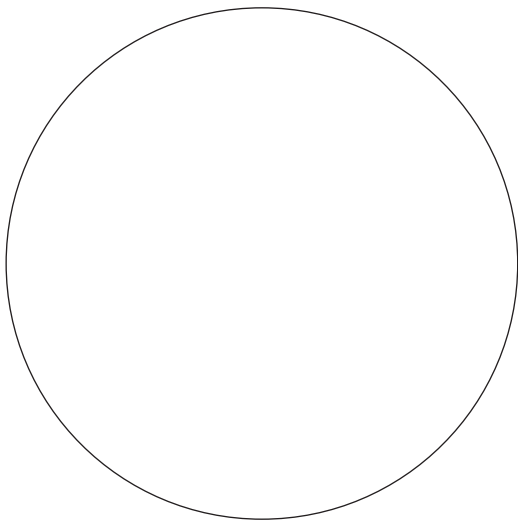
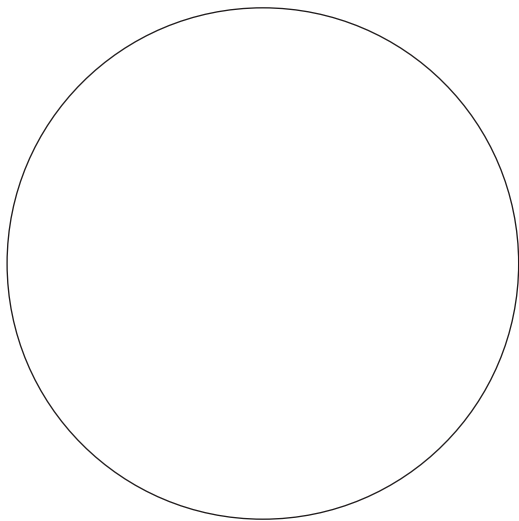
Farms in Australia with land showing signs of salinity

- 2 Use the following data to create two pie charts. Remember to title and label your pie charts correctly.

Irrigated farms showing signs of salinity	
• Dairy – 1 005	• Beef and/or sheep – 698
• Grapevines – 504	• Grain – 411
• Mixed, crop–livestock – 364	• Vegetables – 349
• Fruit – 217	• Other irrigated – 501
Total: 4 049	

Non-irrigated farms showing signs of salinity	
• Beef and/or sheep – 5 396	
• Grain – 4 692	
• Mixed, crop–livestock – 4 578	
• Other non irrigated – 864	
Total: 15 530	

Source: Australian Bureau of Statistics



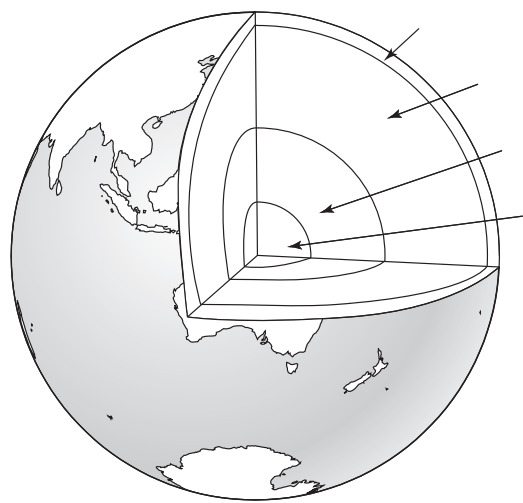
- 3 Answer these questions in your workbook.
- a Are there more irrigated grain farms or non-irrigated grain farms affected by salinity?
 - b What conclusions can you draw from these pie charts?
 - c In what ways could this data be misleading?
 - d How do you think a farm is defined? How could this affect the conclusions that you can draw from this data?
 - e Should a farm be defined by size or income? Or by other criteria? Explain your answer.
 - f How do you think irrigation is defined?
 - g Is the amount of irrigation an important factor when drawing conclusions from this data? Why?

NATURAL HAZARDS 1

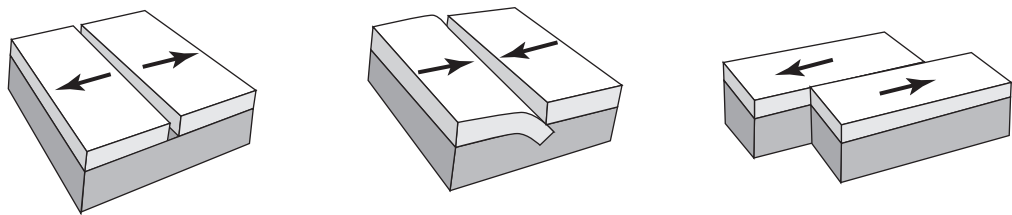
1 Complete the table with examples of Australian and overseas natural hazards that you know of. You may need to share your class knowledge and research places or dates.

Type of natural hazard	Where	When
Earthquake		
Tsunami		2004
	New Orleans	
	Dairy farms in Victoria	1997
	Contamination of water supply	1998
Drought		
	Hong Kong, China, Singapore, Canada	
	Hawaiian Islands	
Fire		

2 Label the four main layers of the Earth's crust, and describe how this relates to your understanding of why earthquakes and volcanoes occur. Annotate the diagram to help explain your answer.



3 a Write the correct type of tectonic movement below each diagram: *convergent*, *transformational*, *divergent*.



b Annotate each diagram to describe the movement occurring and the effects of this movement.

NATURAL HAZARDS 2

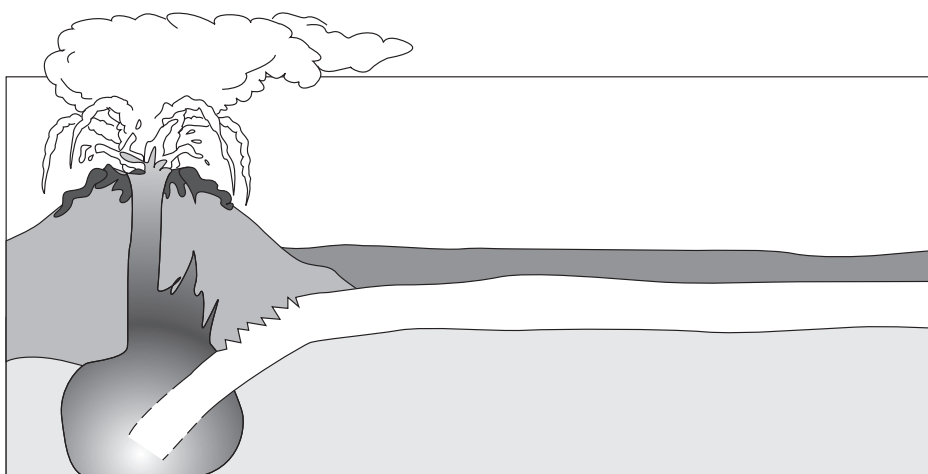
Tectonics quiz

- 4** Use these words to complete the sentences: *fault, millions, crust, move, destructive, oceans, cooled*.

The Earth's _____ is made up of tectonic plates. The Earth's continents and _____ float on these plates. Scientists believe the plates formed many _____ of years ago when the Earth _____. Because earthquakes and tsunamis can occur when the tectonic plates _____, it is the areas near the _____ lines separating plates that mostly experience their effects.

Parts of a volcano

- 5** Label the parts of the volcano using these words: *ash cloud, continental plate, crater, Earth's crust, lava flow, magma chamber, oceanic plate, subduction zone, volcano*.



- b** Draw on the diagram and write short descriptions to show how volcanic eruptions could affect the environment and the people in the area surrounding it.
- 6** Write each hazard beside the correct description: *Ash Wednesday Bushfires, Cyclone Tracy, Hobart Bushfires, Sydney Hailstorm, Newcastle Earthquake, Thredbo Landslide, Meckering Earthquake, Great Influenza Epidemic, Southern Australian Heatwave*.
- | | |
|---|---|
| <p>a I was a strong hazard of my type, but caused little damage due to my remote, sparsely settled location (1968).</p> <p>_____</p> | <p>f In 1967 and 1982, I proved that being our 'cool capital' was no protection from me.</p> <p>_____</p> |
| <p>b I caused extensive damage and death across two Australian states in 1983.</p> <p>_____</p> | <p>g Building regulations in Darwin were changed after my visit in 1974.</p> <p>_____</p> |
| <p>c One person survived me, but I killed 18 others.</p> <p>_____</p> | <p>h In 1939, I killed 438 people (mostly the young and the old), without doing much physical damage at all.</p> <p>_____</p> |
| <p>d I am a biological hazard that killed thousands of people in the winter of 1918.</p> <p>_____</p> | <p>i In January 1994, I arrived at a large, east-coast city in a severe storm that put quite a dent in insurance company profits.</p> <p>_____</p> |
| <p>e I killed 13 people and caused extensive damage to an industrial city in late 1989.</p> <p>_____</p> | |

NATURAL HAZARDS 3

Using the information from the *Natural hazards* slide presentation, as well as what you already know about the 2004 Indian Ocean Tsunami, plan a 300-word news report for the press, radio, television or Internet. Use the following headings as well as your own to guide your planning.

After you have completed the plan, further research the topic and include the additional material to improve your report. Then write out the revised draft.

Prepare a presentation of your report – record it, film it or create an illustrated newspaper column.

Location

Place

What you are doing

Date and time

Weather conditions

Job title

Your reporter name

What you see around you

Who you talk to and what they say

Sounds you hear

Australia's Physical Geography

Worksheets 1–2

- 1** Gradation: weathering, erosion, landslide, deposition. Tectonic: Folding, faulting, vulcanism
- 2** (In row order valley, erosional; delta, depositional; canyon, erosional; gibber plain, residual/erosional; plain, residual/depositional; beach, depositional; mountain, residual; dune, depositional)
- 3** c
- 4** c
- 5** b
- 6** (In order) block mountains, dissected plateau, volcano, fold mountains
- 7a** Largest
- b** flattest
- c** Driest
- 8a** False
- b** True
- c** False
- d** True
- e** False
- 9a** Ningaloo Reef
- b** Kangaroo Island
- c** Fraser Island
- d** Macdonnell Ranges
- e** Wave Rock
- f** Lake Mungo
- g** Daintree Rainforest
- h** Cradle Mountain
- 10a** Tropical Wet-Dry, Monsoon
- b** Tropical East Coast
- c** Tropical Desert
- d** Mid Latitude Continental
- e** Warm Temperate West Coast (Mediterranean)
- f** Warm Temperate East Coast
- g** Cool Temperate West Coast
- 11** (In order) driest, water, deposition, alluvial, pressure, rock, tectonic, exposed, sand
- 12** Teacher to check

Rain Formation

Worksheets 1–2

- 1** Teacher to check. Answers may include: drinking, washing clothes, bathing/showers, industry, agriculture
- 2** Precipitation: Any form of H₂O that falls from the sky, e.g. rain, sleet, snow, hail
- 3** In colder environments, such as the two poles
- 4** Local conditions apply to only a small area, such as localised fog, light showers
- 5** (In order) 5, 1, 3, 4, 2, 6
- 6** (In order) 4, 1, 5, 2, 3
- 7** (In order) 2, 6, 5, 1, 4, 3
- 8a** Rises
- b** Water
- c** Warm
- d** Cools
- e** Releases
- f** Cooling
- g** Rains
- 9a** True
- b** False
- c** True
- d** False
- e** True

Worksheet 3

- 1** C. It is on the windward slope and at the foot of the mountain. It is the location most likely to get more rainfall and runoff
- 2** 1
- 3** Village 1: (higher rainfall) vegetables, timber, dairy produce
Village 2: (lower rainfall) wheat, olives

4–8 Teacher to check

Water and Drainage

Worksheets 1–2

- 1** Liquid – water, solid – ice, gaseous – fog/steam
- 2** Teacher to check
- 3** 1: source, 2: waterfall, 3: tributary, 4: lake, 5: meander, 6: flood plain,

7: estuary, 8: delta, 9: mouth, 10: sea

- 4a** The percentage of water use versus population for each state
- b** Water use increases with population increase
- c** QLD, NSW
- d** Industrial water use, agricultural water use, poorer water saving and water conservation awareness, ratio of land size to population
- e** ACT
- 5a** The average daily household water use of certain countries
- b** Litres used per person
- c** Teacher to check
- d** (Largest to smallest) United States 430 L, Canada 345 L, Australia 340 L, United Kingdom 200 L, Sweden 200 L, South Africa 150 L, Germany 150 L, France 110 L, Israel 100 L, Bangladesh 40 L
- e** The higher the socio-economic level of the country, the higher the water use tends to be, although geographic size could also be a factor
- f** Teacher to check

Worksheet 3

- 1** Teacher to check

Soils and Vegetation

Worksheets 1–2

- 1** Soil or vegetation profiles are cross-sections of a particular area that are often sketched, showing layers of that area that would otherwise be invisible on other types of maps
Geographers use them to understand the profile and features of particular areas of land when surveying them. representing data – for later analysis
- 2** Teacher to check
- 3a** A diagram that shows the hidden parts of an area
- b** A section of rainforest

SELECTED ANSWERS (CONT'D)

- c** The measurements act as a scale, which allows the tree heights to be estimated
- 4a** The rainforest floor receives very little sunlight and is quite humid. Only certain plants can grow here due to low light
- b** The understorey or shrub layer of a rainforest is rich in life, but still sees only minimal sunlight. Large animals inhabit this area and the leaves are large
- c** The forest canopy houses the largest and oldest trees in the rainforest. Biodiversity here is enormous, and it is sometimes said that rainforest canopies could house almost half of the world's life-forms
- d** The emergent layer of the rainforest describes the highest layer of the environment, where trees can grow up to 70 m tall and need to be able to withstand harsh weather conditions. Some species of bats, monkeys and butterflies can be found up here
- 5** Soil is the interface between the lithosphere and the biosphere. It is mainly comprised of precipitate, worn down rocks and minerals
- 6** (In row order)
Weathered rock pieces, limestone, chalk, sandstone
Organic matter, garden scraps, leaves, dead insects
Minerals, carbon, iron, coal
Gases, nitrogen, carbon monoxide, nitrous oxide – via fertilisers applied to soils
Water (what is the source?), underground stream, flooding, domestic garden watering
Living organisms, worms, beetles, slaters
- 7** (In row order)
Past and present climate: Australia drifts north into warmer zones. The climate became drier at the end of the last ice age.
Impacts of inhabitats: Growing aridity and fire stick practices

drop the biodiversity numbers of native flora, introduced species of flora arrive. Aborigines enter the continent and practice fire-stick farming, which encourages fire-resistant species.
Europeans then settle the continent, begin clearing and introduce exotic species
Forest protection programs: Following the Rio Earth Summit, Australia begins a program of Regional Forest Agreements to protect our forests and the communities economically dependent on them

8 Teacher to check

9a 62.3%

b 37.7%

10a 71.3%

b 28.3%

Worksheet 3

- 1** (In order) poor, salinity, evaporation, irrigation, underground, minerals, topsoil, farming
- 2** Teacher to check
- 3** Teacher to check

Natural Hazards

Worksheets 1-2

- 1** Examples include (in row order):
Tsunami: Indian Ocean, 2004
Cyclone: New Orleans
Anthrax: Dairy farms in Victoria, 1997
Contamination of water supply: Sydney, 1998
Drought: Australia-wide – various dates and main areas impacted, 1895–1902 (teacher to check)
SARS: Hong Kong, China, Singapore, Canada, 2003
Volcano: Hawaiian Islands, 1996
- 2** Labels should be (from inside out) inner core, outer core, mantle, crust. Teacher to check annotations

- 3** (In order) divergent, convergent, transformation
- 4** (In order) crust, oceans, millions, cooled, move, fault
- 5** Teacher to check
- 6a** Meckering earthquake
- b** Ash Wednesday fires
- c** Thredbo landslide
- d** Great Influenza Epidemic
- e** Newcastle earthquake
- f** Hobart fires
- g** Cyclone Tracy
- h** Southern Australian heatwave
- i** Sydney hailstorm

Worksheet 3

- 1** Teacher to check

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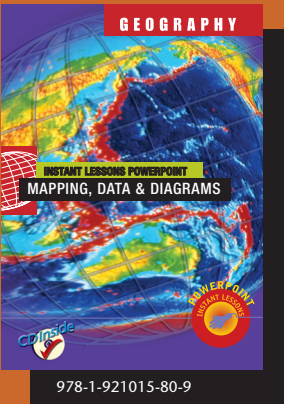
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Animated diagrams and full-colour illustrations	✓	Worksheet tasks and assessments in hard copy and on CD	✓
Key definitions and concepts are explained and illustrated	✓	Suitable for Windows and Mac users	✓

Our Physical World

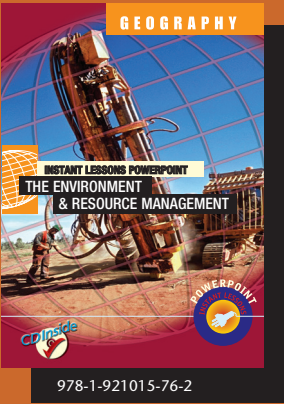
This title of five slide sets and the accompanying notes and worksheets focus on physical geography topics as they relate to Australia and its regional neighbours. The unique characteristics of Australian landforms that were developed through gradational and tectonic forces are explored, as well as regional differences and weather patterns. Frontal, orographic and convectional rain formation types are explained in a series of animations. Slides about the water cycle, water-use patterns and the pressure of population on resources provide an up-to-date review of this important topic. The characteristics of local vegetation, soil profiles, soil fertility, salinity and their impact on agriculture are featured, as well as the impacts of natural hazards such as earthquakes, volcanoes, tsunamis, fire, flood, extreme weather events and disease epidemics.

Slide sets	
1: Australia's Physical Geography 2: Rain Formation 3: Water and Drainage	4: Soils and Vegetation 5: Natural Hazards

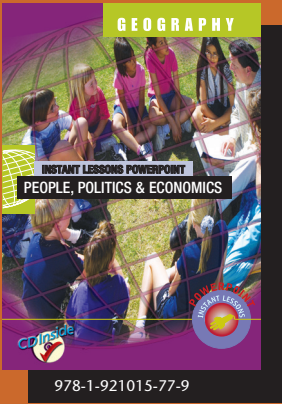
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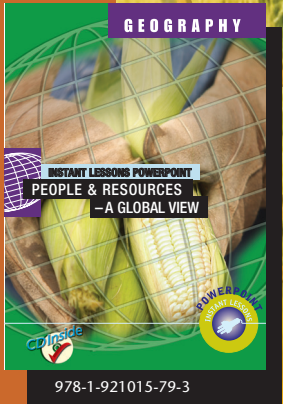
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